Gustavo Almeida Silva

Statistician | Data & Analytics

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Professional Summary

Statistician (B.Sc., Federal University of Juiz de Fora) and Data & Analytics Analyst at Itaú Unibanco, with experience in epidemiology, biostatistics experimentation and, currently, Customer Experience analytics in the financial sector.

At Itaú Unibanco since 2024, with rotations through Operational Risk and Itaú BBA (the bank's wholesale segment) Products. Currently part of the Data Intelligence team within the Customer Service community, applying statistical analysis and modeling to customers experiencing friction with the bank.

Specialized in highly imbalanced modeling problems, predicting customer contacts in friction scenarios before they occur, with a base rate of roughly 1 customer per 100,000 service interactions. Also builds the datasets that map the journey of a customer in friction, using change point detection models to capture atypical events.

Professional Experience

Itaú Unibanco — Junior Data & Analytics Analyst

Customer Service Directorate | Oct 2025 – Present

- Statistician on the Data Intelligence and Critical Flows team within the Customer Service community, leading work on customer journeys in friction and critical flows.
- Developed Machine Learning models (LightGBM and XGBoost) for early identification and prediction of customers in friction with the bank.
- Conducted journey simulation and modeling studies using stochastic processes, including Markov Chains and Hidden Markov Models (HMM), as well as financial optimization projections using generalized time series models.
- Built data pipelines to identify friction moments during service interactions and to generate features from those events for use in predictive models.
- Worked directly with regulators and industry bodies, such as BACEN and FEBRABAN, on the reporting and validation of official Customer Service (SAC) and Institutional Ombudsman data.

Itaú BBA — Intern II

Cash Management PJ Directorate | Jul 2025 – Sep 2025

- Member of the BUPJ Products team, working within the Data Solutions for Clients team.
- Developed hyper-personalized Market Intelligence reports for retail companies, using GenAI architectures built with LangGraph.
- Developed solutions to recommend new franchise openings in the Wholesale segment, using geo-referenced modeling and clustering algorithms applied to competing companies.

Itaú Unibanco — Intern I

Corporate Banking (PJ) Planning Directorate | Aug 2024 – Jul 2025

- Member of the BUPJ Products team, working in Operational Risk Management.
- Developed dashboards (Excel, Power Query and SQL) for risk event monitoring and analysis.
- Created communications and materials to engage and train business teams on the risk agenda.
- Planned and monitored the Operational Risk Score OKR for Itaú Empresas.

CNPq — Undergraduate Research Fellow (Scientific Initiation)

Aug 2023 – Aug 2024

- Built pipelines for data collection, modeling and automated deployment of epidemiological forecasting models (R Targets, Sparklyr and Tidymodels).
- Developed the FableCount package for predictive modeling of count time series using the Fable framework in R.
- Actively engaged with the epidemiology scientific community in the Zona da Mata region (MG), supporting several municipalities in data-driven decision-making for COVID-19 and other respiratory diseases.

UFJF – Jf Salvando Todos — Undergraduate Research Fellow (Scientific Initiation)

Jun 2021 – Aug 2023

- Developed automated epidemiological data analysis reports.
- Designed and modeled databases (SQL).
- Built interactive data visualizations.

Statistics Consultant for Graduate Students

- Provided statistical consulting to Master's and Ph.D. students across multiple fields of knowledge.
- Covered everything from descriptive and exploratory analyses to complex sample design, design of experiments and causal modeling.
- Key projects: public policy evaluation in the state of Alagoas using Causal Inference methods; Box-Behnken experimental design to optimize milk derivatives for protein production (partnership with BIOQUIMTEC – UFJF).

Education

Bachelor of Science in Statistics — Federal University of Juiz de Fora (UFJF) | 2020 – 2025

Open-Source Projects

- **fableCount** (R package) — Author. Modeling of count time series.
- **MMCSD** (R package) — Co-author. Modeling of complex longitudinal survey data.

Certifications

- Gen-AI Itaú Associate Certified — Itaú IOX, 2025
- Data & Analytics Engineering Itaú Associate Certified — Itaú IOX, 2026

Languages

Portuguese (Native) | English (Fluent) | Spanish (Basic)